1 Approximate Integration

In the last lesson, we saw that if $f : [a,b] \to \mathbb{R}$ is integrable, then

$$\lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x = \int_a^b f(x)dx,$$

where we choose a uniformly-spaced partition P of width $\Delta x = \frac{b-a}{n}$ and $\tau = \{x_i^*\}_{i=1}^n$ is any tagging of P. Given an error tolerance $\varepsilon > 0$, our goal is to find the smallest positive integer n such that

$$\left| \sum_{i=1}^{n} f(x_i^*)\Delta x - \int_a^b f(x)dx \right| < \varepsilon.$$

The Riemann sum $\sum_{i=1}^n f(x_i^*)\Delta x$ is our desired approximation of $\int_a^b f(x)dx$. Different choices of tagging $\tau = \{x_i^*\}_{i=1}^n$ affect whether our approximation is an over- or under-estimate, as well as the rate of convergence. We explore this issues in the next section.

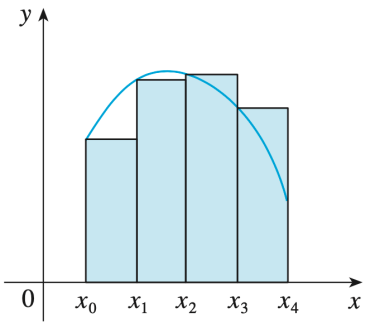
2 Left-endpoint, right-endpoint, and midpoint approximations

Definition 2.1. The following are common choices of tagging τ when approximating a definite integral:

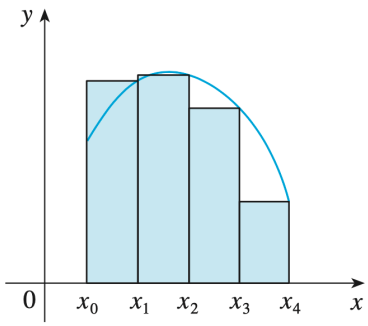
(i) The *left-endpoint approximation* is given by taking $x_i^* = x_{i-1}$ for each $i = 1,\ldots,n$.

(ii) The *right-endpoint approximation* is given by taking $x_i^* = x_i$ for each $i = 1,\ldots,n$.

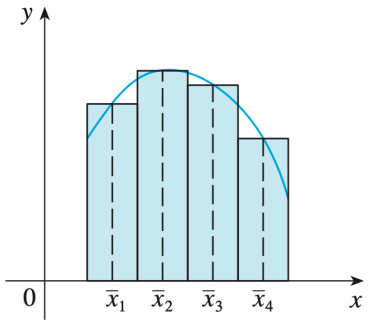
(iii) The *midpoint approximation* is given by taking $x_i^* = \frac{x_{i-1}+x_i}{2}$ for each $i = 1,\ldots,n$.



(a) Left endpoint approximation



(b) Right endpoint approximation



(c) Midpoint approximation

1. Use the left-endpoint, right-endpoint, and midpoint approximations with $n = 5$ to approximate $\int_1^2 \frac{1}{x} dx$ to 6 decimal places. Compare each approximation to the exact answer $\ln(2) \approx 0.693147$ by calculating the error $\sum_{i=1}^{n} f(x_i^*)\Delta x - \ln(2)$.

3 Error Bounds

Suppose we approximate $\int_0^1 e^{x^2} dx$ using one of the methods above. Since we don't know the exact value of the integral, we also don't know the error in our approximation, so how do we know if our approximation is any good? While it is not possible to determine the exact error, the theorems below show that in many cases we can at least find an upper bound on the absolute error.

Theorem 3.1 (Error bound for left- and right-endpoint approximations)**.** *Let E_n^L and E_n^R denote the absolute error $|\sum_{i=1}^n f(x_i^*)\Delta x - \int_a^b f(x)dx|$ using the left- and right-endpoint approximations, respectively. If $M = \max\{|f'(x)|\}_{x\in[a,b]}$, then E_n^L and E_n^R are both at most $M\frac{(b-a)^2}{2n}$.*

Theorem 3.2 (Error bound for the midpoint approximations)**.** *Let E_n^M denote the absolute error $|\sum_{i=1}^n f(x_i^*)\Delta x - \int_a^b f(x)dx|$ using the midpoint approximation. If $M = \max\{|f''(x)|\}_{x\in[a,b]}$, then $E_n^M \leq M\frac{(b-a)^3}{24n^2}$.*

Before proving these, we will get some practice using them.

4 Calculating error bounds

2. Calculate the error bounds for your approximations in Problem 1, and compare them to the actual error.

3. How large should we take n in order to guarantee that the Midpoint approximation for $\int_1^2 \frac{1}{x} dx$ is accurate to within 0.0001?

4. (a) Use the Midpoint Rule with $n = 10$ to approximate the integral $\int_0^1 e^{x^2} dx$.

   (b) Given an upper bound for the error in this approximation.

   (c) How large should we take n in order to guarantee that the Midpoint Approximation for $\int_0^1 e^{x^2} dx$ is accurate to within 0.0001?